Observation

TYPE OF CELL	COLOUR OF CYTOPLASM	COLOUR OF NUCLEUS	COLOUR OF GRANULES
RBC	pink	-	-
WBCs(leucocytes)			
Neutrophil	pink	blue	lilac
Eosinophil	pink	blue	orange
Basophil	pink	blue	Dark blue black
lymphocyte	blue	violet	-

Results

The blood smear was stained using field's stain and erythrocytes and leucocytes were observed under microscope.

5. Test for Catalase

Aim

To test whether the given culture is catalase positive by the catalase test

Theory And Principle

Catalase test demonstrates the presence of catalase, an enzyme that catalyses the release of oxygen from hydrogen peroxide (H_2O_2). It is used to differentiate those bacteria that produces an enzyme catalase, such as *staphylococci*, from non-catalase producing bacteria such as *streptococci*.

The enzyme catalase mediates the breakdown of hydrogen peroxide into oxygen and water. The presence of the enzyme in a bacterial isolate is evident when a small inoculum is introduced into hydrogen peroxide, and the rapid elaboration of oxygen bubbles occurs. The lack of catalase is evident by a lack of or weak bubble production. The culture should not be more than 24 hours old.

Requirements

- Slides
- Nichrome loop or toothpick
- 24hour old culture
- 3%hydrogen peroxide
- Dropper

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